

10 (**short**) A particle moves in a spiral trajectory defined by constant angular velocity $\dot{\theta} = \Omega$ and radial position given by $r = r_0 e^{\beta t}$, where r_0 and β are positive constants and t is time.

(a) Calculate the velocity and acceleration of the particle in polar coordinates. [6]

(b) For what combination of parameters does the radial component of the acceleration equal 0? [4]